

Chain of Thought Prompting for Large Language Model Reasoning

Hard Language Tasks: Reasoning

Reasoning Problems

Q: If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?

A: The answer is **5**

Arithmetic Reasoning (AR)
(+ - × ÷ ...)

Q: Take the last letters of the words in "Elon Musk" and concatenate them

A: The answer is **nk**.

Symbolic Reasoning (SR)

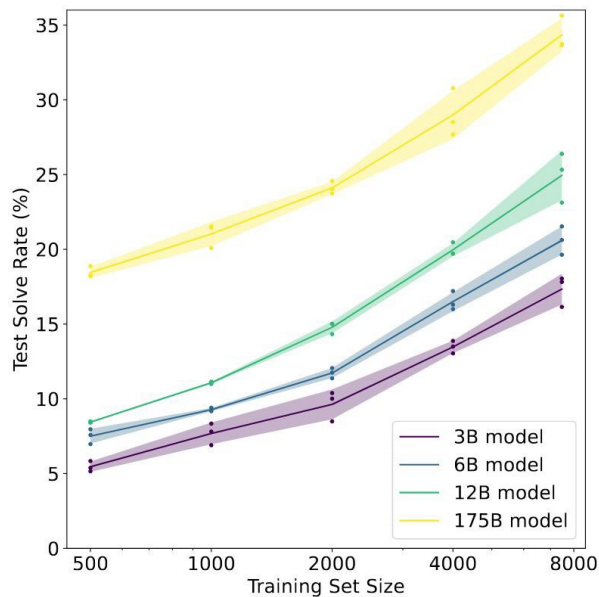
Q: What home entertainment equipment requires cable?
Answer Choices: (a) radio shack (b) substation (c) television (d) cabinet

A: The answer is **(c)**.

Commonsense Reasoning (CR)

Reasoning Problems

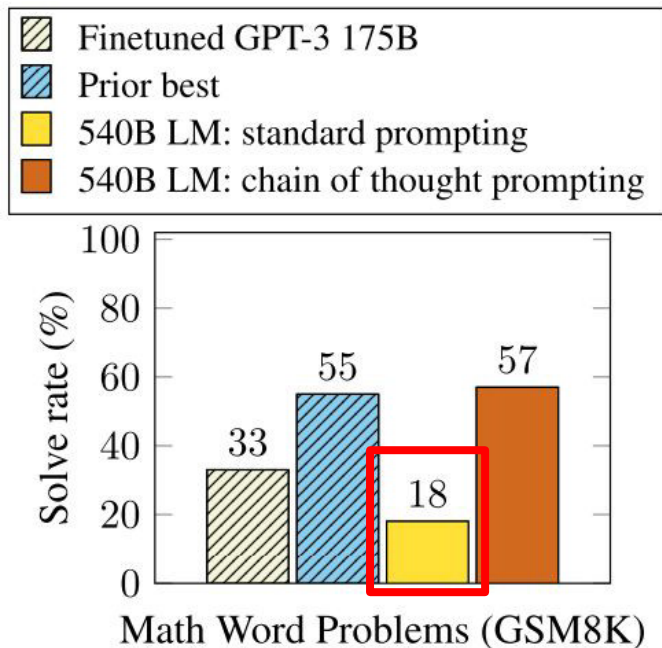
Fine-tune GPT-3 on GSM8K (arithmetic): (Cobbe et al. 2021)



Conjecture: to achieve $> 80\%$, needs 100 times more fine-tuning data for 175B model

Reasoning Problems

GSM8K (arithmetic):



Few-shot standard prompting with even larger model (PaLM 540B) also does not work well.

Reasoning Problems

Scaling up language model size does not **efficiently** achieve high performances, for Arithmetic Reasoning (*AR*), CommonSense Reasoning (*CR*) and Symbolic Reasoning (*SR*) tasks.

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Proposed solution: **chain of thought prompting**

Chain of Thought Prompting

Chain of Thought (CoT)

Chain of Thought Prompting Elicits Reasoning in Large Language Models

Few-Shot CoT

**Jason Wei Xuezhi Wang Dale Schuurmans Maarten Bosma
Brian Ichter Fei Xia Ed H. Chi Quoc V. Le Denny Zhou**

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Both papers will appear
in **NeurIPS'22!**

Large Language Models are Zero-Shot Reasoners

Zero-Shot CoT

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Chain of Thought (CoT)

Definition:

A chain of thought is **a series of intermediate natural language reasoning steps** that lead to the final output.

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Intuition (from neural-symbolic computing):

use <input, **intermediate results**, output> triples, rather than simple <input, output> pairs

Benefits:

- Decomposition -> easier intermediate problems
- Interpretable
- More general than neural symbolic computing
- Leveraging prompting of LLM

Chain of Thought (CoT)

(a) Few-shot

Examples

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ❌

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(b) Few-shot-CoT (Wei et al., 2022)

CoT Examples

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. **✓**

Step-by-step Answer

Chain of Thought (CoT)

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(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

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(Output) The answer is 8. **X**

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CoT Examples

Step-by-step Answer

(d) Zero-shot-CoT (KoJima et al., 2022)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

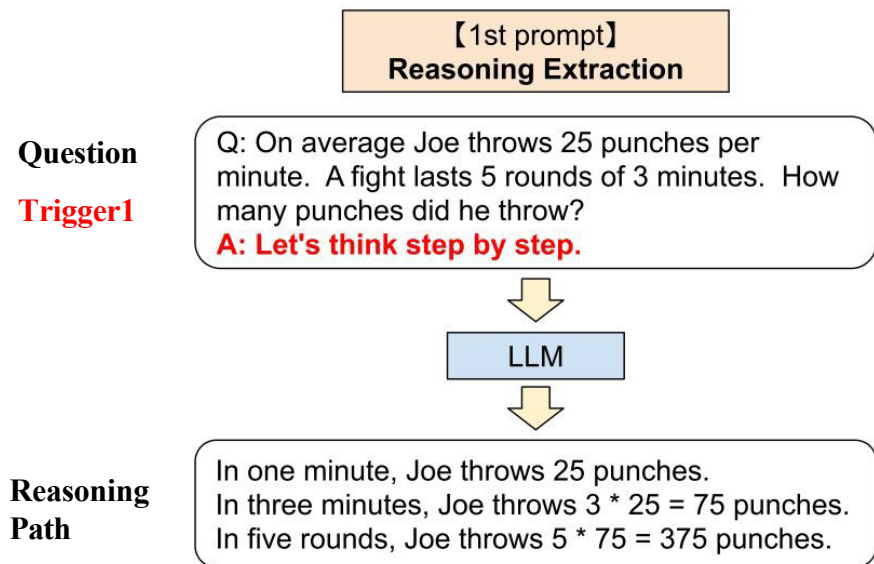
A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. **✓**

Two-stage Prompting Step-by-step Answer

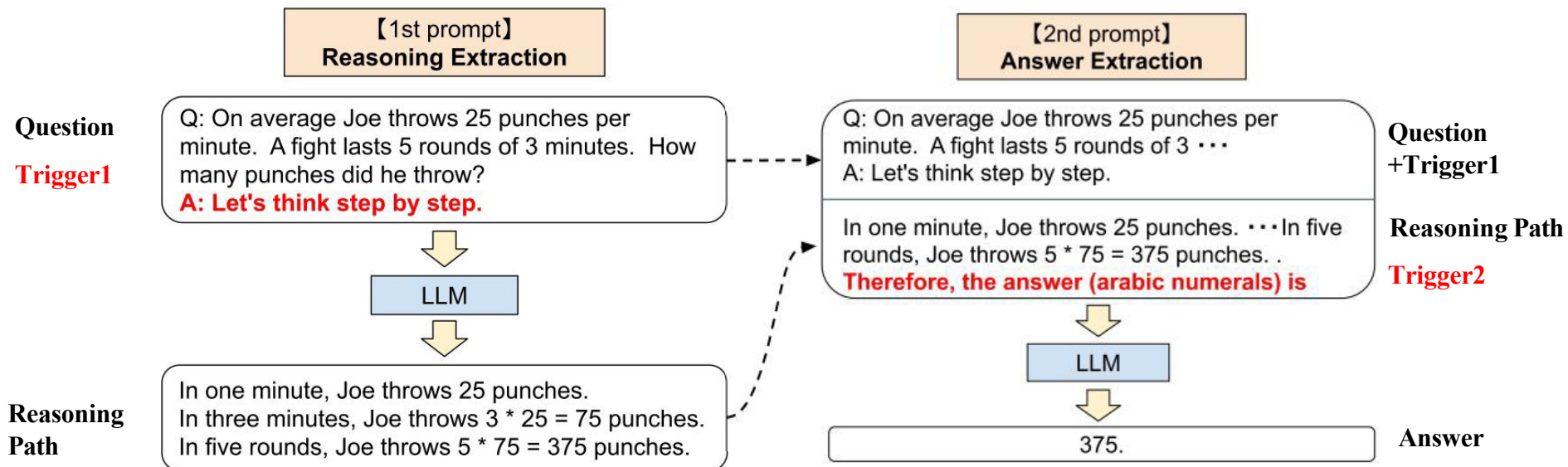
Zero-Shot Chain of Thought (CoT)

For **zero-shot CoT**, a **two-stage prompting** is applied:



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Experiments

Models

Pre-trained LLMs:

- Instruct GPT-3 (ada 350M, babbage 1.3B, curie 6.7B, and davinci 175B) (Ouyang et al., 2022)

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- Instruct GPT-3 (ada 350M, babbage 1.3B, curie 6.7B, and davinci 175B) (Ouyang et al., 2022)
 - Not your familiar GPT-3 (Brown, et al., 2020)
 - Fine-tuned with human feedback
 - Stay tuned for the lecture on **Nov. 14!!**

Models

Pre-trained LLMs:

- Instruct GPT-3 (ada 350M, babbage 1.3B, curie 6.7B, and davinci 175B) (Ouyang et al., 2022)
- PaLM (8B, 62B, 540B) (Chowdhery et al., 2022)
 - Only accessible to Googlers 🙄.

Models

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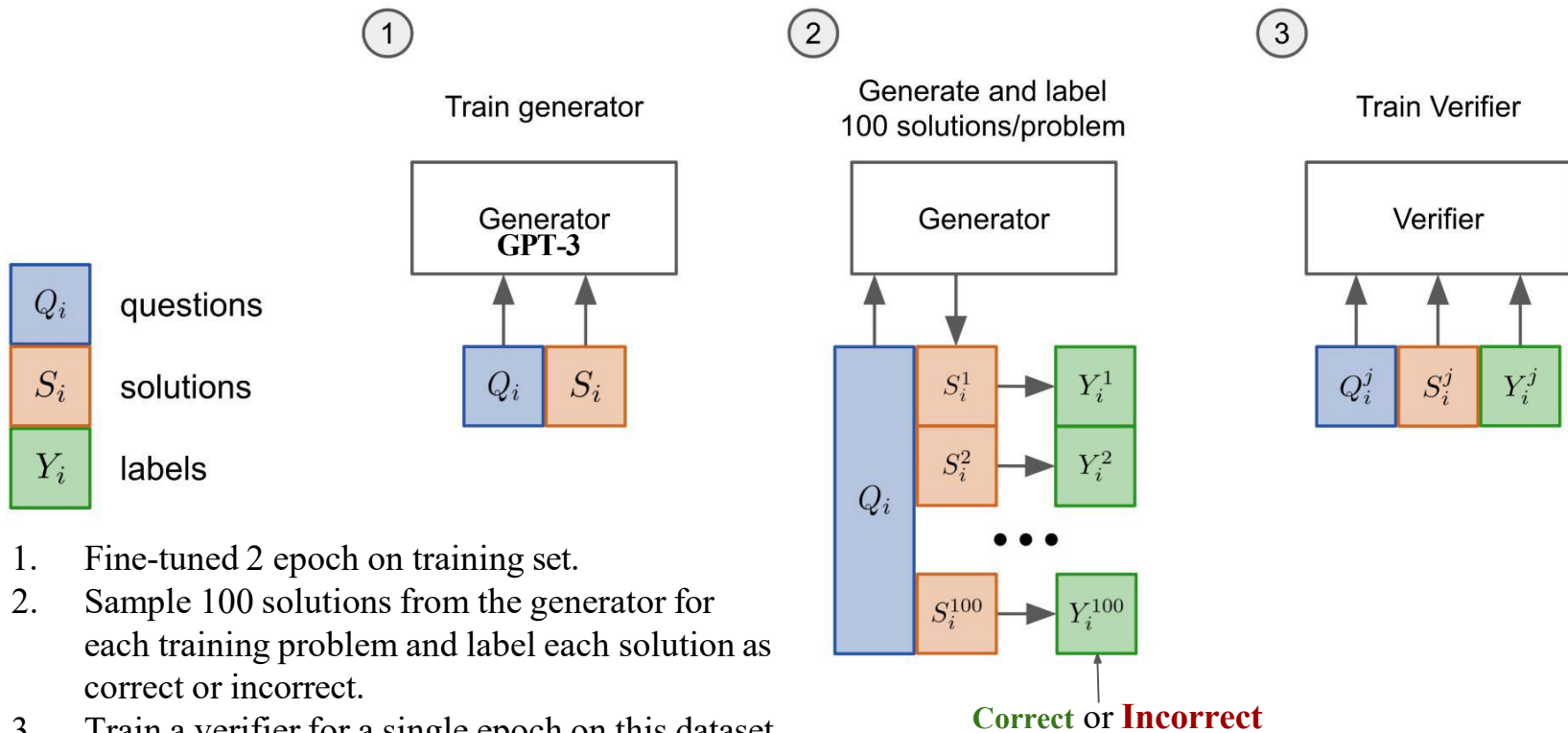
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- PaLM (8B, 62B, 540B) (Chowdhery et al., 2022)
- LaMDA (422M, 2B, 8B, 68B, 137B) (Thoppilan et al., 2022)
 - Dialogue-oriented LM.
 - Fine-tuned on human-annotated data.

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- GPT-3 (ada 350M, babbage 1.3B, curie 6.7B, davinci 175B)
- GPT-2 (1.5B)
- GPT-Neo (2.7B), GPT-J (6B), T0 (11B) (Sanh et al., 2022), OPT (13B) (Zhang et al., 2022)

Prior Best – Fine-tuning + Verification



Experiments

Arithmetic Reasoning

Free Response - Few-Shot CoT Prompt Exemplar

Free Response

Q: If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?

A: **There are originally 3 cars. 2 more cars arrive. $3 + 2 = 5$.** The answer is **5**.

(Wei et al., 2022)

- **Manually** composed 8 exemplars

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Free Response

Q: Olivia has \$23. She bought five bagels for \$3 each. How much money does she have left?

A: **Olivia had 23 dollars. 5 bagels for 3 dollars each will be $5 \times 3 = 15$ dollars. So she has $23 - 15$ dollars left. $23 - 15$ is 8. The answer is 8.**

(Wei et al., 2022)

You can have one or more equations.

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Equations can be incomplete and combined math with words.

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- **Manually** composed **8** exemplars
- All contains equations with flexible formats
- Benchmarked on:
 - **GSM8K** (Cobbe et al. 2021)
 - **SVAMP** (Patel et al., 2021)
 - **MAWPS** (Koncel-Kedziorski et al., 2016)

Multiple Choice - Few-Shot CoT Prompt Exemplar

Multiple Choice

Q: A person is traveling at 20 km/hr and reached his destiny in 2.5 hr then find the distance?

Answer Choices: (a) 53 km (b) 55 km (c) 52 km (d) 60 km (e) 50 km

A: **The distance that the person traveled would have been 20 km/hr * 2.5 hrs = 50 km.** The answer is **(e)**.

- 4 exemplars, whose questions, intermediate reasoning, and answers are from AQuA-RAT's **training set**

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Multiple Choice

Q: If $a / b = 3/4$ and $8a + 5b = 22$, then find the value of a.

Answer Choices: (a) $1/2$ (b) $3/2$ (c) $5/2$ (d) $4/2$ (e) $7/2$

A: **If $a / b = 3/4$, then $b = 4a / 3$. So $8a + 5(4a / 3) = 22$. This simplifies to $8a + 20a / 3 = 22$, which means $44a / 3 = 22$. So a is equal to $3/2$. The answer is (b).**

The exemplars have **various** formats.

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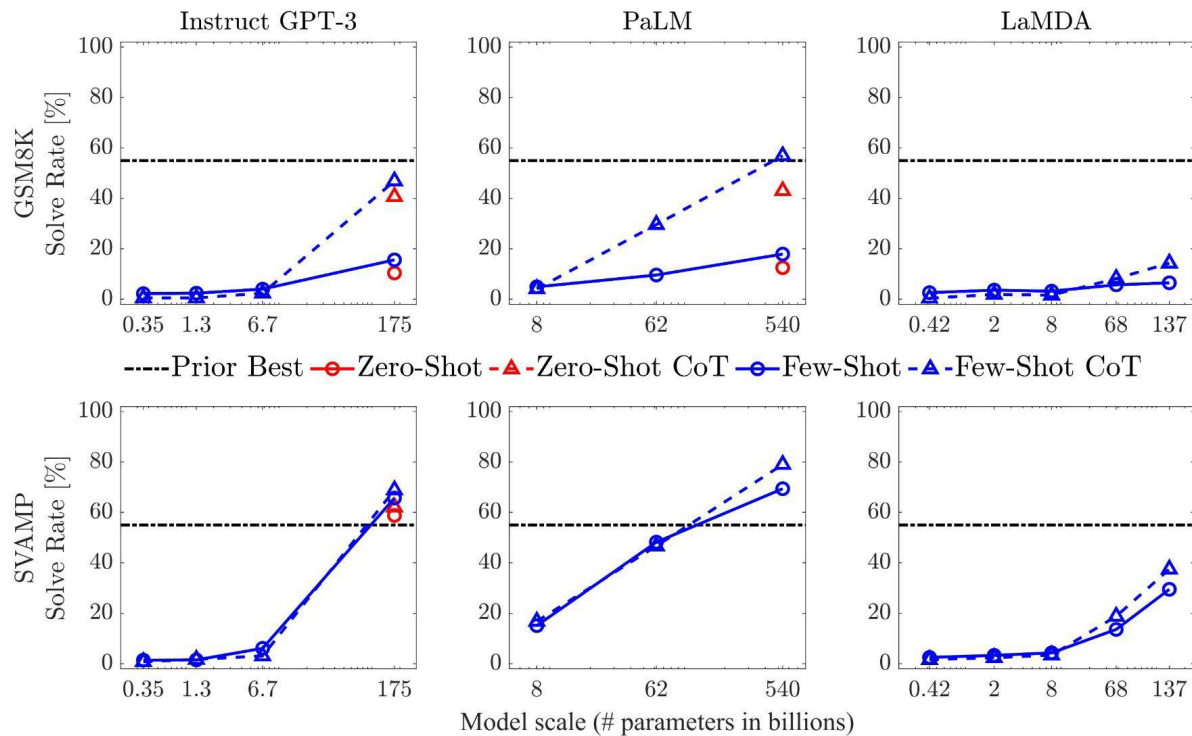
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- Benchmarked on **AQuA-RAT (Ling et al., 2017)**

Arithmetic Reasoning - Results



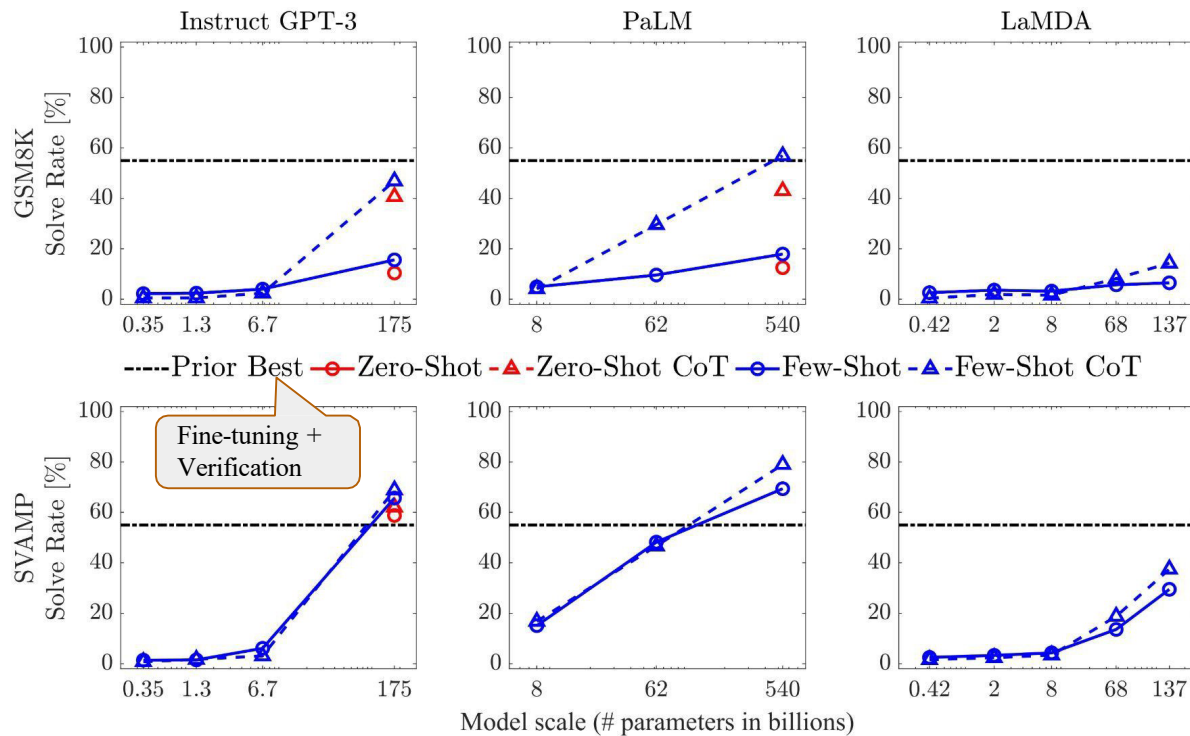
GSM8K

Josh decides to try flipping a house. He buys a house for \$80,000 and then puts in \$50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

SVAMP

Each pack of dvds costs 76 dollars. If there is a discount of 25 dollars on each pack. How much do you have to pay to buy each pack?

Arithmetic Reasoning - Results



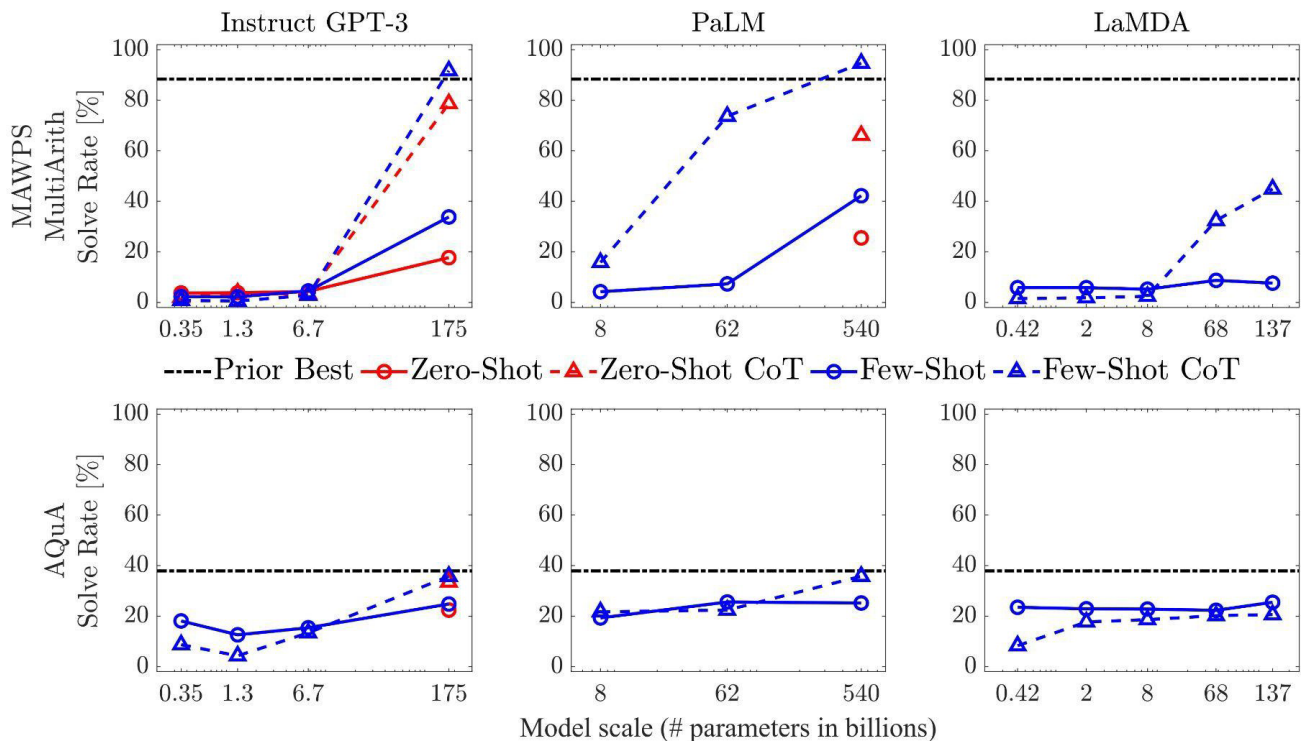
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MAWPS - MultiArith

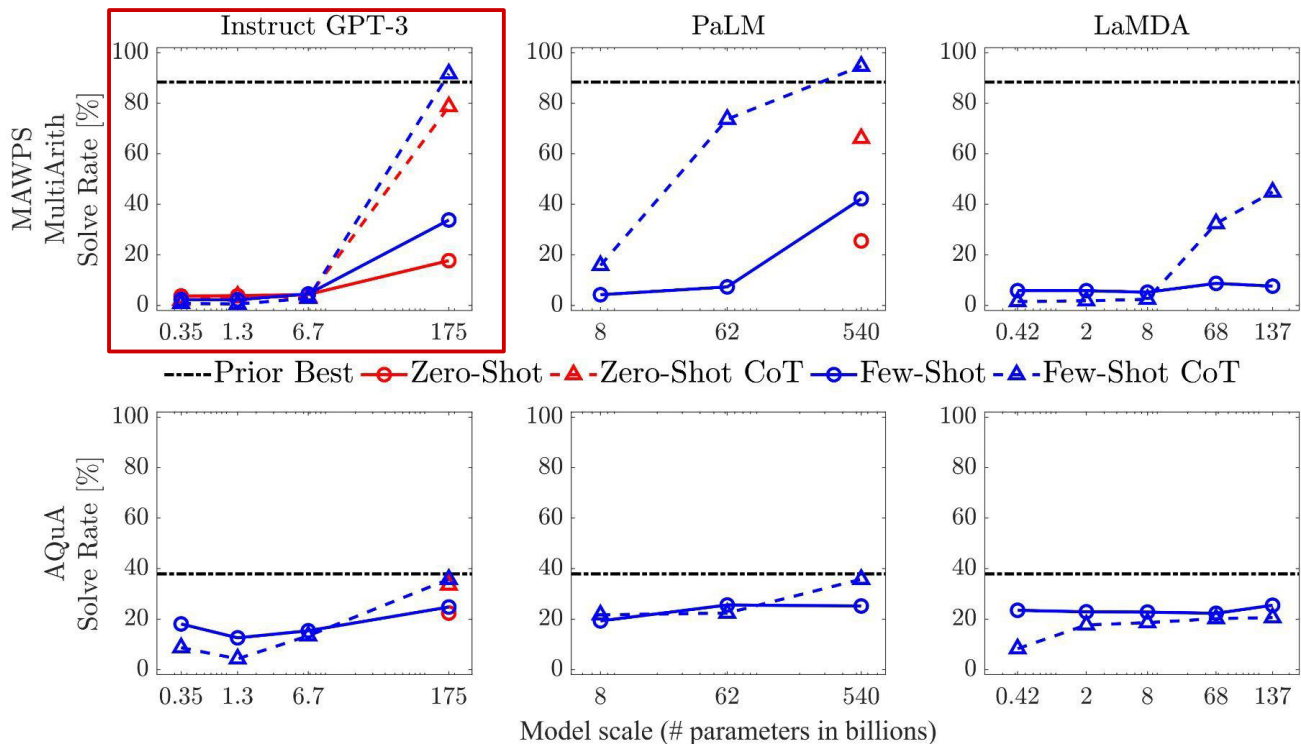
The school cafeteria ordered 42 red apples and 7 green apples for students lunches. But, if only 9 students wanted fruit, how many extra did the cafeteria end up with?

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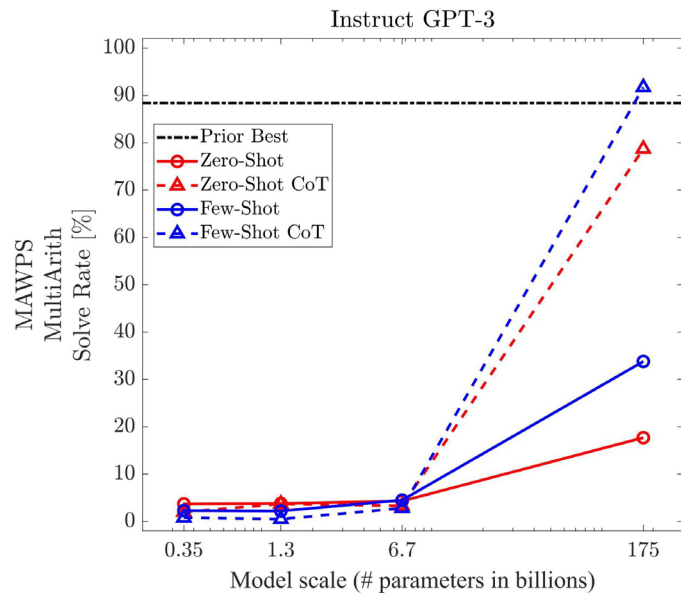
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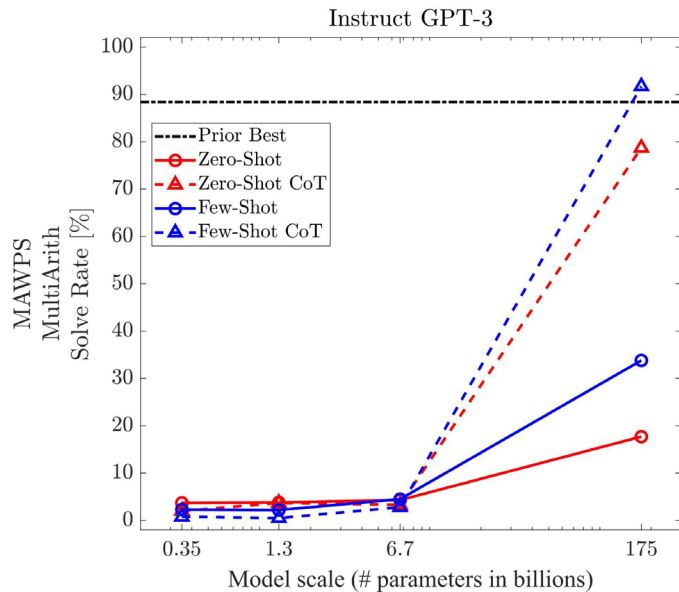
Arithmetic Reasoning - Observations

- Both zero-shot and few-shot chain of thought promptings are emergent **abilities of model scale**.



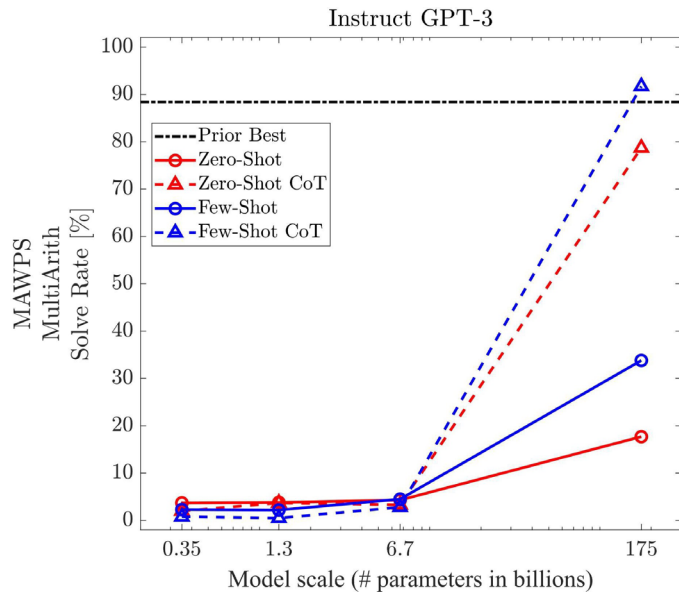
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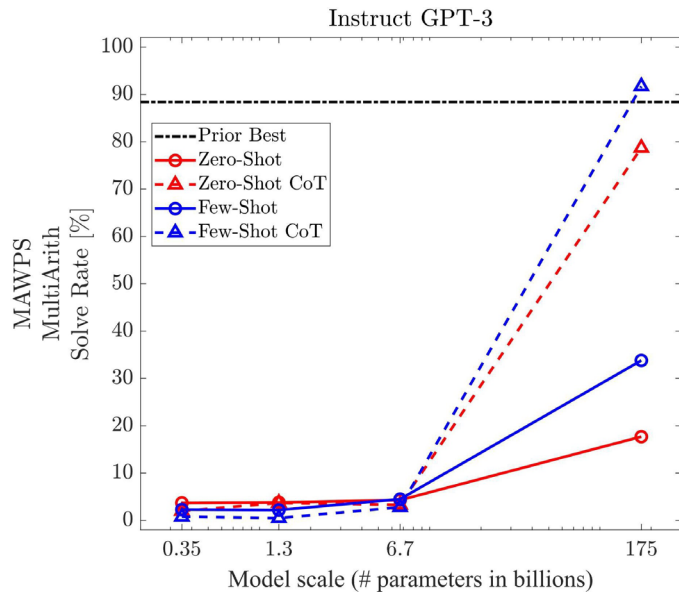
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- Few-shot CoT achieves **better** performance on LLM than zero-shot CoT.
- Instruct GPT-3: text-davinci-002 achieves similar performance as PaLM 540B model



Experiments

Symbolic Reasoning

Symbolic Reasoning - Last Letter Concatenation

Last letter concatenation

Q: Take the last letters of the words in "Elon Musk" and concatenate them

A: The last letter of "Elon" is "n".
The last letter of "Musk" is "k".
Concatenating them is "nk". The answer is **nk**.

- Generate full names by **randomly concatenating** names from the **top one-thousand first and last** names from name census data

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- 4 exemplars with **strict** format

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Last letter concatenation

Q: Take the last letters of the words in "Bill Gates" and concatenate them

A: The last letter of "Bill" is "l".
The last letter of "Gates" is "s".
Concatenating them is "ls". The answer is **ls**.

Symbolic Reasoning - Coin Flip

Coin Flip

Q: A coin is heads up. Tom does not flip the coin. Mike does not flip the coin. Is the coin still heads up?

A: The coin was flipped by no one. So the coin was flipped 0 times. The coin started heads up, and it was not flipped, so it is still heads up. So the answer is yes.

Coin Flip

Q: A coin is heads up. Jamey flips the coin. Teresa flips the coin. Is the coin still heads up?

A: The coin was flipped by Jamey and Teresa. So the coin was flipped 2 times, which is an even number. The coin started heads up, so after an even number of flips, it will still be heads up. So the answer is yes.

8 exemplars with **strict** format.

Symbolic Reasoning - In & Out-of-domain Test

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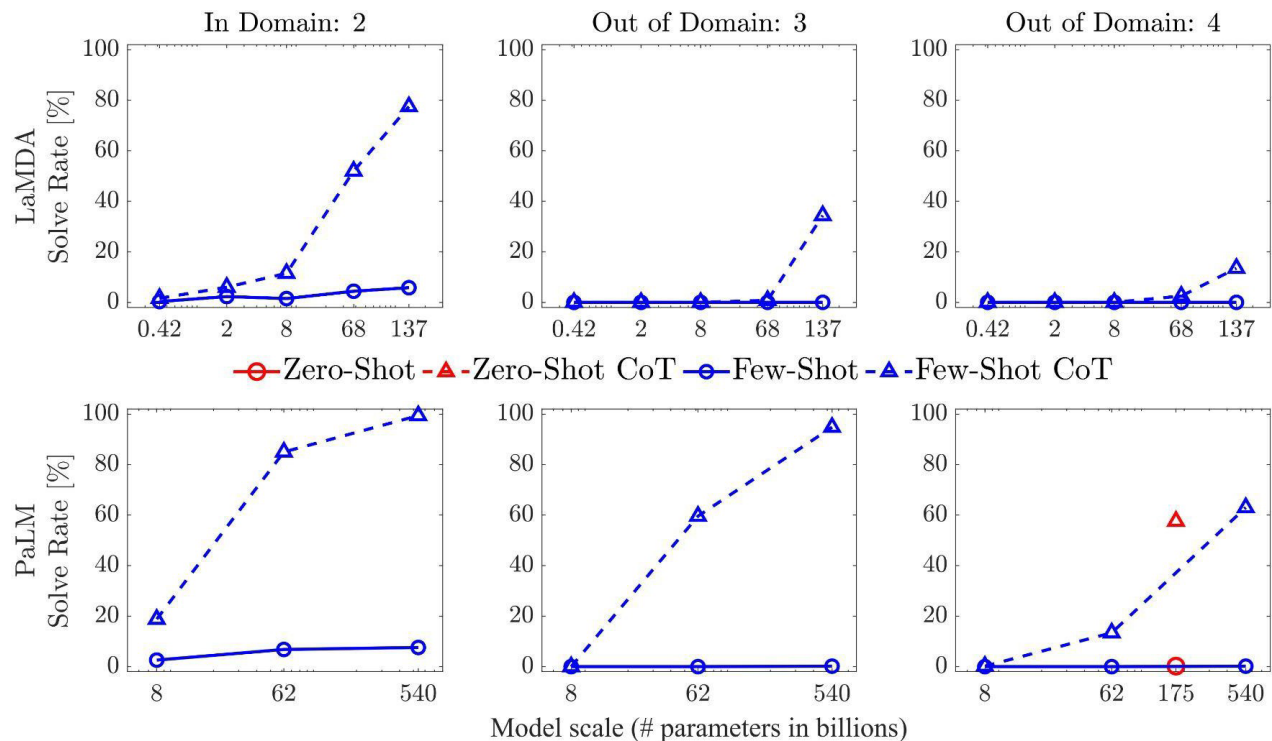
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A: **The coin was flipped by no one. So the coin was flipped 0 times. The coin started heads up, and it was not flipped, so it is still heads up.** So the answer is **yes**.

- In-domain test set: examples had the same number of steps as the few-shot exemplars
- Out-of-domain (OOD) test set: examples had more steps than those in the exemplars.

Symbolic Reasoning - Last Letter Concatenation



In-Domain

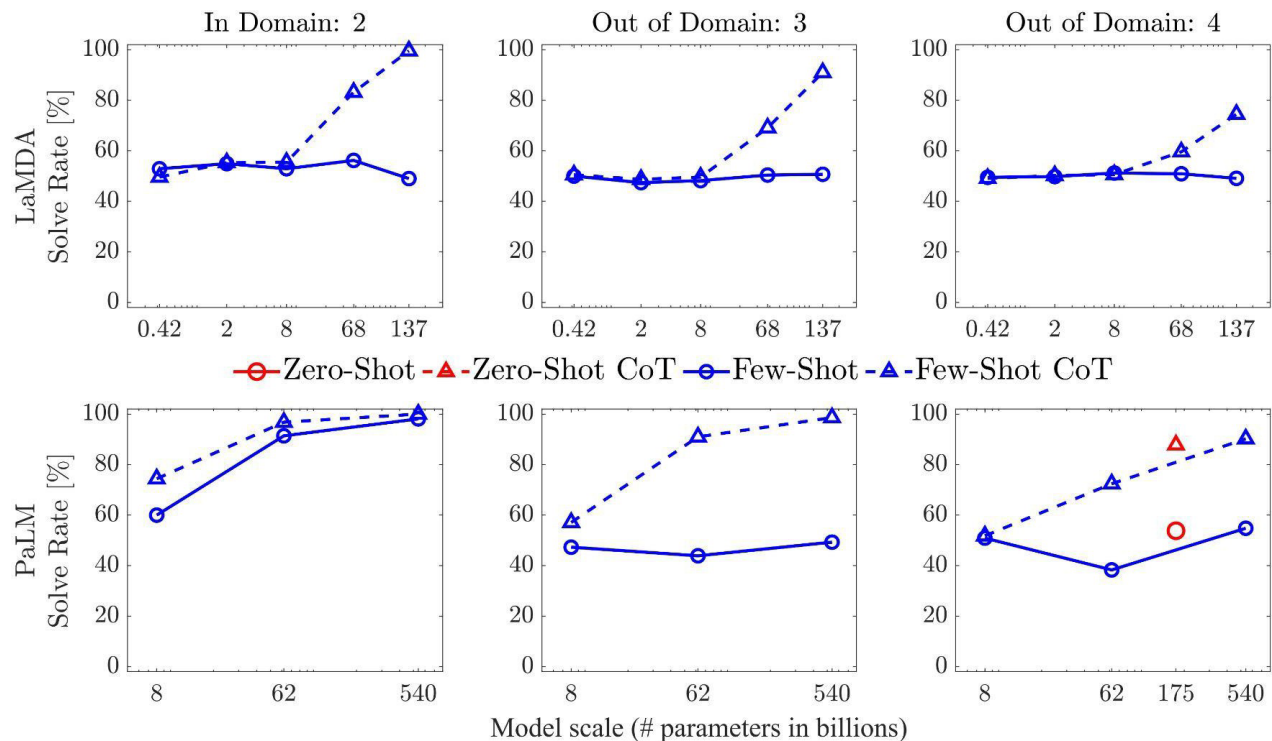
Take the last letters of the words in "**Elon Musk**" and concatenate them.

Out-of-Domain

Take the last letters of the words in "**Johann Sebastian Bach**" and concatenate them.

*Zero-Shot results use **Instruct-GPT-3 175B** text-davinci-002 model.

Symbolic Reasoning - Coin Flip



In-Domain

A coin is heads up. **Tom does not flip the coin. Mike does not flip the coin.** Is the coin still heads up?

Out-of-Domain

A coin is heads up. **Tom does not flip the coin. Mike does not flip the coin. Jake flips the coin.** Is the coin still heads up?

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Symbolic Reasoning - Observations

- Standard prompting **fails out-of-domain** tests for both tasks.
- Few-shot CoT prompting is **emergent abilities of model scale**.
- CoT does not positively impact performance for small models, but start to yield performance gains when using models with more than ~100B parameters for both in-domain and out-of-domain tests.
- **Zero-shot** CoT using **Instruct-GPT-3 175B** achieves the similar performance as **few-shot** CoT in both tasks using **540B PaLM model**.

Pre-Lecture Question 2

Q2: Wei et al., 2022 showed that CoT can improve out-of-domain performance. Can you state their results and why do you think it is the case (i.e., adding intermediate steps can improve robustness)?

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While standard prompting fails out-of-domain tests for both tasks, large models with both zero-shot and few-shot CoT improve the performance of in-domain and out-of-domain tests. CoT prompt in this symbolic reasoning tasks really guides the LM to reason the process of mapping input to the output. Even if questions are *OOD* in the sense of “*how many words in the name*” and “*how many states to track*”, the process of getting the output are still same, and can be resembled from exemplars. However, it is still unsure that CoT will improve other *OOD* scenarios with more complex reasoning processes.

Experiments

CommonSense Reasoning

Commonsense Reasoning - Toy Problems

CSQA (Talmor et al., 2019)

Q: What home entertainment equipment requires cable?

Answer Choices: (a) radio shack (b) substation (c) television (d) cabinet

A: The answer is **(c)**.

StrategyQA (Geva et al., 2021)

Q: Could Brooke Shields succeed at University of Pennsylvania?

A: The answer is **yes**.

Sport Understanding

Q: Is the following sentence plausible? "Jamel Murray was perfect from the line."

A: The answer is **yes**.

Date Understanding

Q: 2015 is coming in 36 hours. What is the date one week from today in MM/DD/YYYY

A: So the answer is **01/05/2015**.

Commonsense Reasoning - Toy Problems

SayCan Robot Planning

Locations = [counter, table, user, trash, bowl].

Objects = [7up, apple, kettle chips, tea, multigrain chips, coke, lime soda, jalapeno chips, rice chips, orange, grapefruit soda, pepsi, redbull, energy bar, sponge, water].

The robot can pick up items with **pick(object)** and put down items with **put(object)** as well as find objects or locations with **find()**. The robot can only understand the explicit locations and objects listed.

Human: How would you throw away a redbull?

Plan: 1. **find(redbull)**, 2. **pick(redbull)**, 3. **find(trash)**, 4. **put(redbull)**, 5. **done()**.

Commonsense Reasoning - Toy Problems

SayCan Robot Planning

Locations = [counter, table, user, trash, bowl].

Objects = [7up, apple, kettle chips, tea, multigrain chips, coke, lime soda, jalapeno chips, rice chips, orange, grapefruit soda, pepsi, redbull, energy bar, sponge, water].

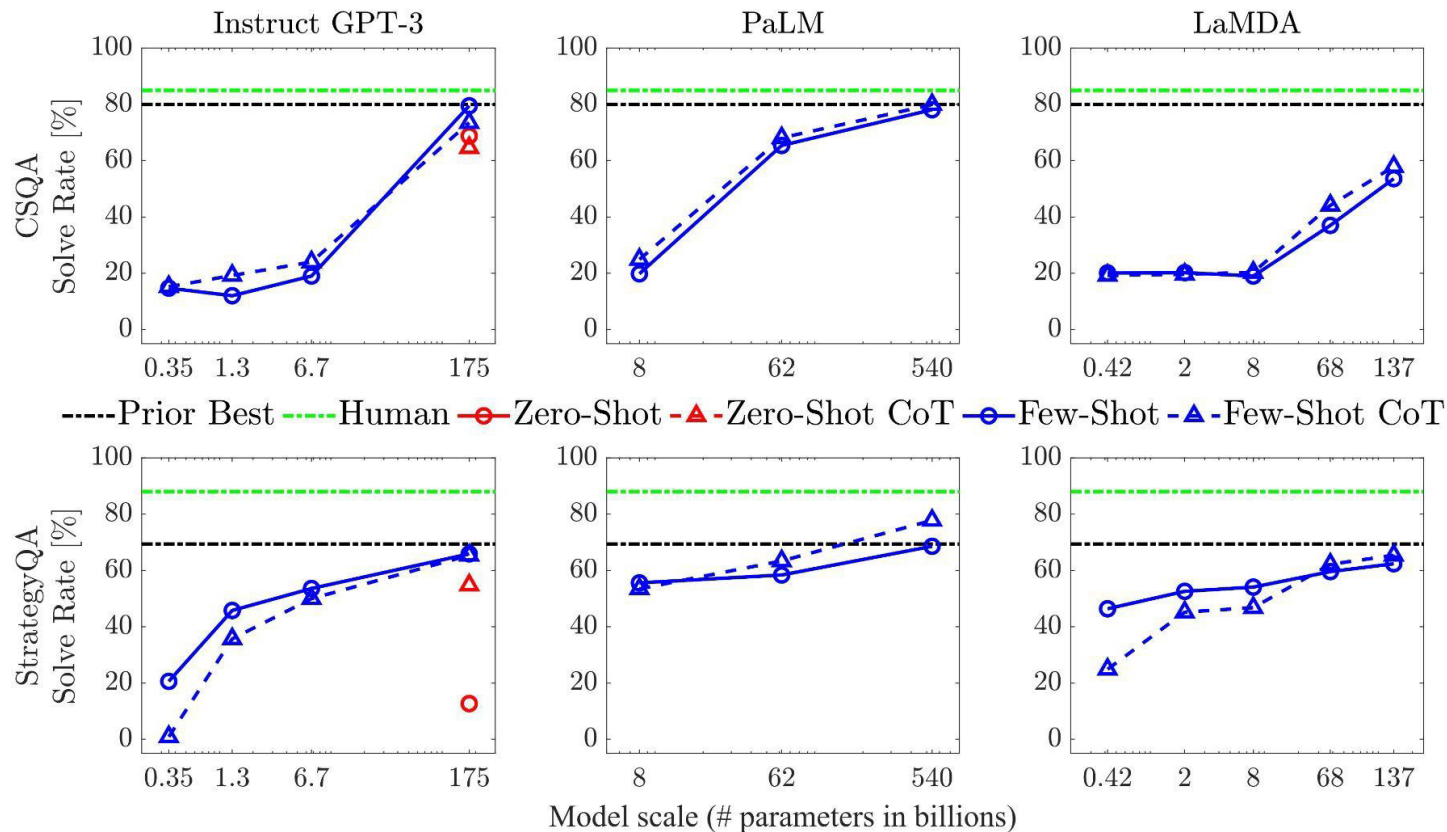
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Human: How would you throw away a redbull?

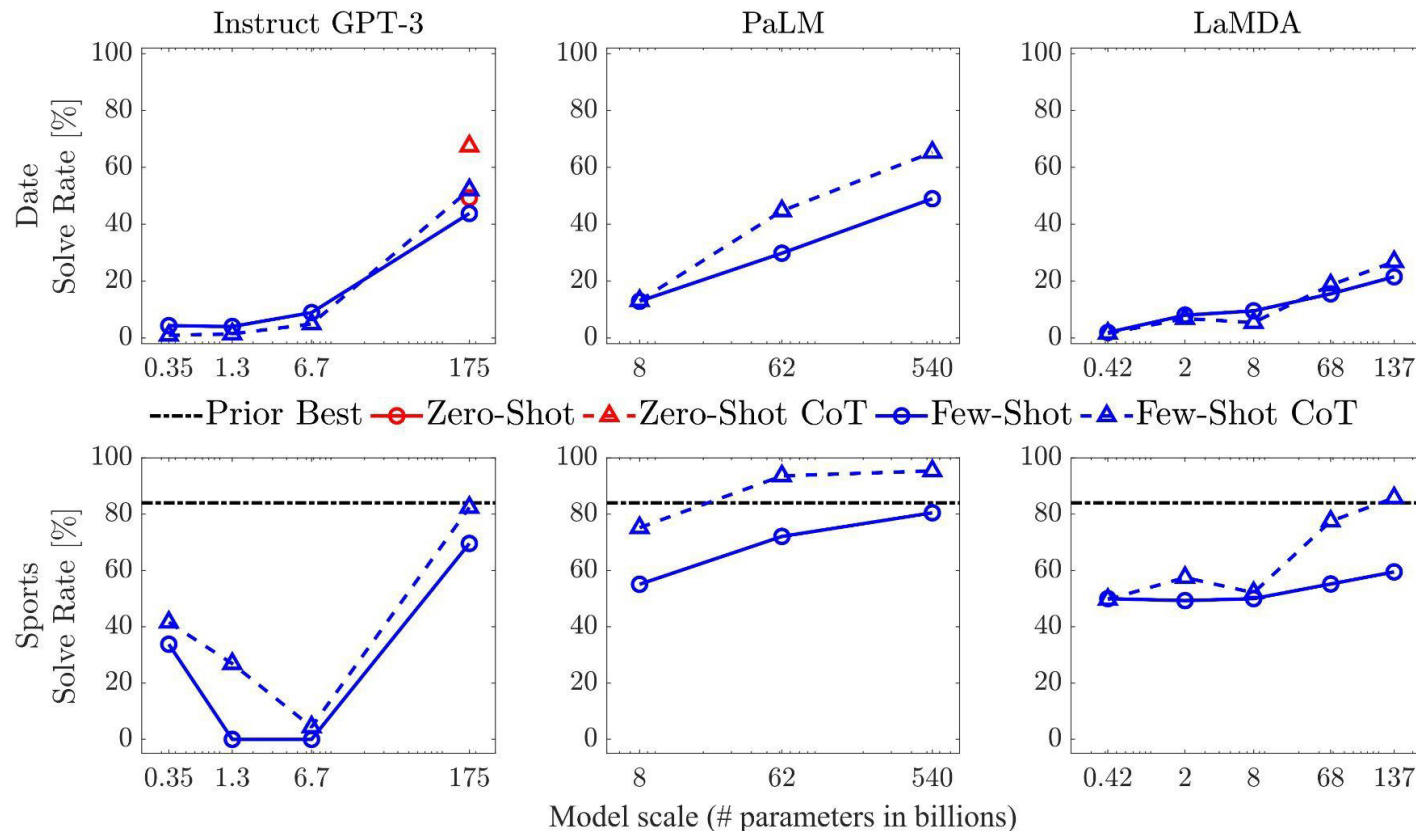
Plan: 1. **find(redbull)**, 2. **pick(redbull)**, 3. **find(trash)**, 4. **put(redbull)**, 5. **done()**.

These tasks not only require **multi-steps reasoning**, but also need **priori knowledge** to understand complex semantics.

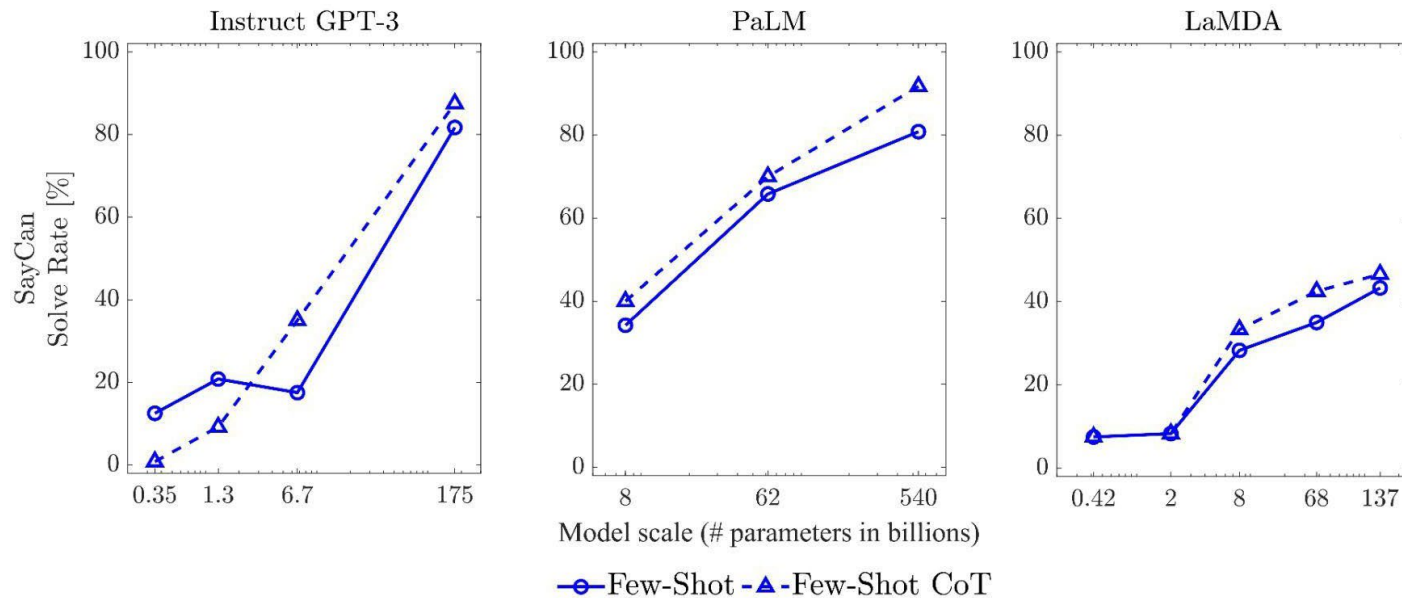
Commonsense Reasoning - Results



Commonsense Reasoning - Results



Commonsense Reasoning - Results



Commonsense Reasoning - Observations

- For all tasks, scaling up **model size** improved the performance of standard prompting.

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Commonsense Reasoning - Observations

- For all tasks, scaling up **model size** improved the performance of standard prompting.
- CoT prompting led to further gains, with improvements appearing to be **largest for PaLM 540B**.
- CoT has **minimal** benefits on CSQA and StrategyQA tasks.
- **Few-shot** achieves better performance than **Zero-shot** CoT on 175B GPT-3 model for CSQA and Strategy QA tasks, but **Zero-shot** CoT shows significant improvement for **Date understanding** task.

Ablation Study - Variations of Few-Shot CoT

Change the types of CoT:

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

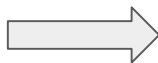
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

Equation only



$5+6=11$. The answer is 11.

Ablation Study - Variations of Few-Shot CoT

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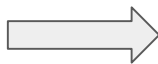
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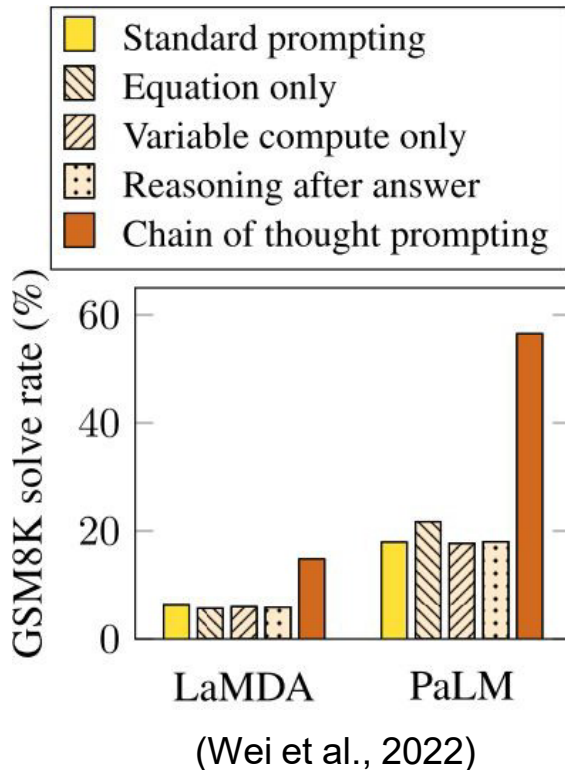
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Natural language in reasoning matters.



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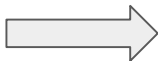
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Variable compute only



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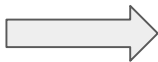
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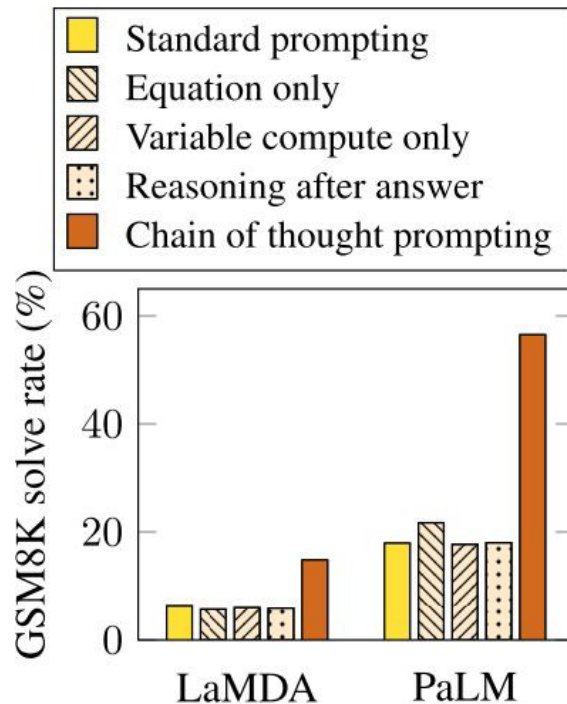
A:

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Variable compute only



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..... The
answer is 11.



More intermediate computation does not help with the final answer.

(Wei et al., 2022)

Ablation Study - Variations of Few-Shot CoT

Change the types of CoT:

(b) Few-shot-CoT

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Reasoning after answer



*The answer is 11.
Roger started with
5 balls. 2 cans of 3
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Ablation Study - Variations of Few-Shot CoT

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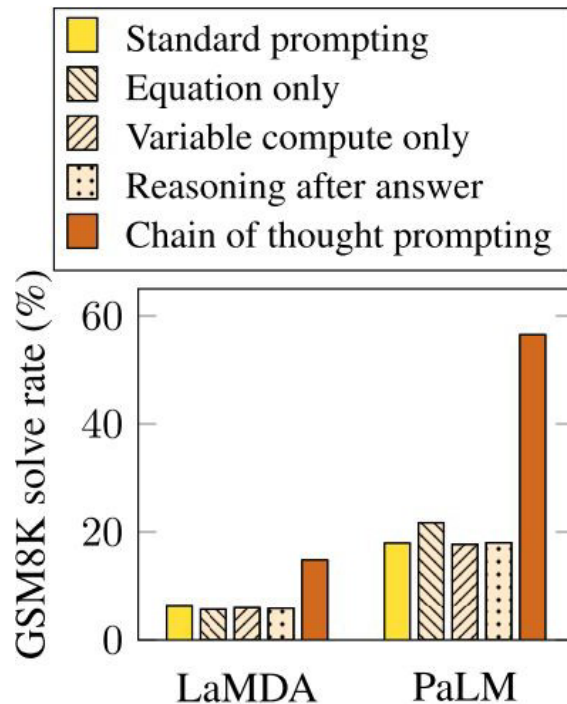
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Reasoning after answer



*The answer is 11.
Roger started with
5 balls. 2 cans of 3
tennis balls each is
6 tennis balls. $5 +$
 $6 = 11$.*



(Wei et al., 2022)

CoT is not just activating knowledge seen in pre-training.

Ablation Study - Robustness (Style of Exemplar)

Change the style of exemplar in **few-shot CoT**:

(b) Few-shot-CoT

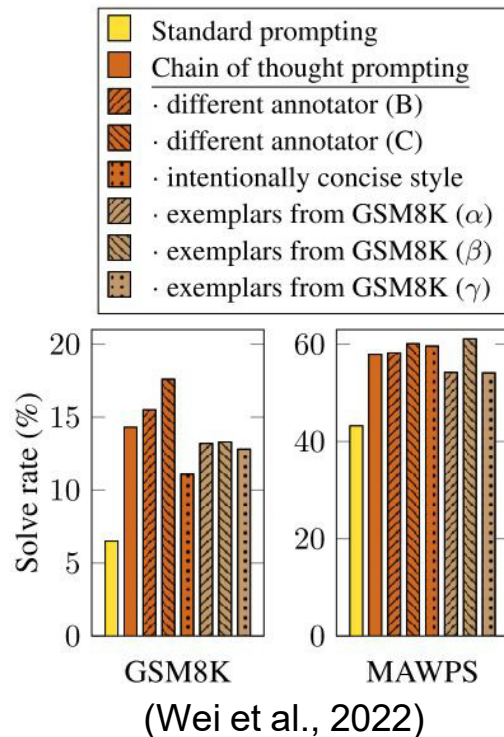
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Results for **few-shot** LaMDA 137B on two **AR** tasks: have variance, but CoT still outperforms standard prompting, **robust against linguistic styles, different exemplars**.

Ablation Study - Robustness (Trigger Sentence)

Change the template (trigger sentence) in
zero-shot CoT:

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) *There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓*

No.	Template	Accuracy
1	Let's think step by step.	78.7
2	First, (*1)	77.3
3	Let's think about this logically.	74.5
4	Let's solve this problem by splitting it into steps. (*2)	72.2
5	Let's be realistic and think step by step.	70.8
6	Let's think like a detective step by step.	70.3
7	Let's think	57.5
8	Before we dive into the answer,	55.7
9	The answer is after the proof.	45.7
-	(Zero-shot)	17.7

(Kojima et al., 2022)

Results for **zero-shot** GPT3 (davinci-002) 175B on MultiArith **AR** task: different templates encourage the model to express reasoning quite differently

Ablation Study - Model Size

Change the model sizes in CoT prompting:

(Kojima et al., 2022)

	Original GPT-3 (0.3B/1.3B/6.7B/175B)		Instruct GPT-3 (0.3B/1.3B/6.7B/175B-1/175B-2)		
Zero-shot	2.0 / 1.3 / 1.5 / 3.3		3.7 / 3.8 / 4.3 / 8.0 / 17.7		
Few-shot	5.2 / 5.2 / 4.0 / 8.1		3.0 / 2.2 / 4.8 / 14.0 / 33.7		
Zero-shot-CoT	1.7 / 2.2 / 2.3 / 19.0		2.0 / 3.7 / 3.3 / 47.8 / 78.7		
Few-shot-CoT	4.3 / 1.8 / 6.3 / 44.3		2.5 / 2.5 / 3.8 / 36.8 / 93.0		
	GPT-2 (1.5B)	GPT-Neo (2.7B)	GPT-J (6B)	T0 (11B)	OPT (13B)
Zero-shot	3.2	3.0	2.7	2.8	3.7
Zero-shot-CoT	2.2	1.3	2.5	3.2	2.2

Results on MultiArith **AR** task with different model sizes:

- Larger model, better reasoning
- CoT is effective only for larger models
- Few-shot better than zero-shot
- Instruct GPT-3 is much better than original GPT-3

Pre-Lecture Question 1

Q1: Describe how the two approaches from (Wei et al., 2022) and (Kojima et al., 2022) are different. Which one do you think is a more viable solution in terms of cost, performance and stability?

Wei's work uses few-shot setting with several demonstration examples required, and the CoT annotations need to be provided for each example, while Kojima's work uses LLM to generate the CoT with the two-stage prompting and no longer requires the annotated examples.

For most of benchmarks we have seen, few-shot CoT has better performance than zero-shot CoT, while zero-shot CoT does not require human annotations, which can be costly.

Although there is no direct comparison between zero-shot and few-shot CoT on stability, few-shot seems to be more robust as its performance does not vary significantly when changing the prompt annotations. On the other hand, zero-shot CoT has significant performance variance with different trigger sentences.

More Advances - Self-Consistency

Change greedy decode (single-path) to self-consistency (multi-path) in few-shot CoT:

Prompt with example chains of thought

Q: If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?

A: There are 3 cars in the parking lot already. 2 more arrive. Now there are $3 + 2 = 5$ cars. The answer is 5.

...

Q: Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder for \$2 per egg. How much does she make every day?

A:

Language
model

Sample decode with
diverse reasoning paths

She has $16 - 3 - 4 = 9$ eggs left. So she makes $\$2 * 9 = \18 per day. **The answer is \$18.**

This means she uses $3 + 4 = 7$ eggs every day. So in total she sells $7 * \$2 = \14 per day. **The answer is \$14.**

She eats 3 for breakfast, so she has $16 - 3 = 13$ left. Then she bakes muffins, so $13 - 4 = 9$ eggs left. She makes $9 * \$2 = \18 . **The answer is \$18.**

Majority vote

The answer is \$18

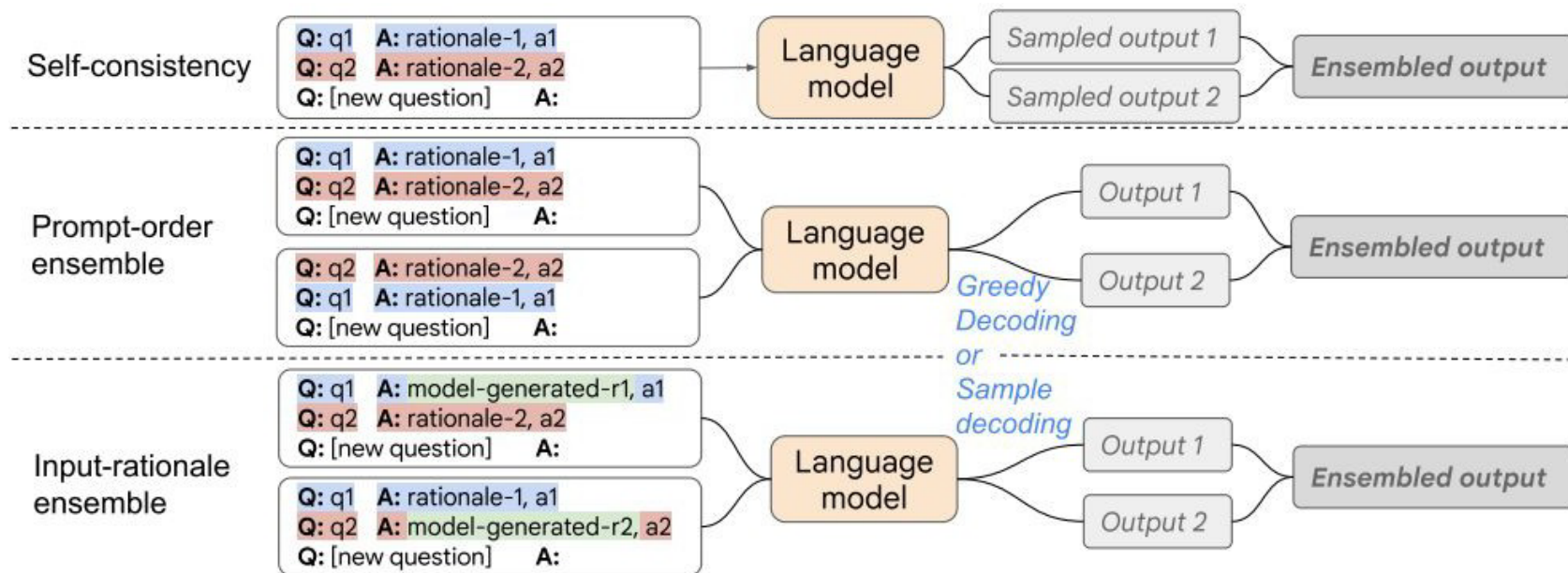
More Advances - Self-Consistency

Showcase results on **AR**, **CR** tasks:

	Method	GSM8K	CommonsenseQA
	Previous SoTA	35 ^e / 57 ^g	91.2^a
LaMDA (137B)	Greedy decode (Single-path)	17.1	57.9
	Self-Consistency (Multi-path)	27.7 (+10.6)	63.1 (+5.2)
PaLM (540B)	Greedy decode (Single-path)	56.5	79.0
	Self-Consistency (Multi-path)	74.4 (+17.9)	80.7 (+1.7)

More Advances - Input-Rational Ensemble

Use model-generated rationale in few-shot CoT:



More Advances - Input-Rational Ensemble

Showcase performance for AR reasoning tasks (PaLM-540B):

	GSM8K
Standard-prompting	17.9
Few-shot CoT (Wei et al. 2022)	56.5
Zero-shot CoT (Kojima et al. 2022)	43.0
Self-consistency (Wang et al. 2022)	74.4
Prompt-order ensemble	75.4
Input-rationale ensemble	73.8

Performance improvement on reasoning is great over previous CoT, but not significant against self-consistency,

Pre-Lecture Question 3 and Discussion

Q3: Do you think the CoT method can be useful to other NLP tasks that we have seen in the previous lectures (standard NLP tasks that are beyond the arithmetic/logic reasoning tasks that these papers evaluated on)? Do you have any ideas about how we can collect the CoT data?

Reference

1. Wei, Jason, et al. "Chain of thought prompting elicits reasoning in large language models." *arXiv preprint arXiv:2201.11903* (2022).
2. Kojima, Takeshi, et al. "Large Language Models are Zero-Shot Reasoners." *arXiv preprint arXiv:2205.11916* (2022).
3. Cobbe, Karl, et al. "Training verifiers to solve math word problems." *arXiv preprint arXiv:2110.14168* (2021).
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5. Miao, Shen-Yun, Chao-Chun Liang, and Keh-Yih Su. "A diverse corpus for evaluating and developing English math word problem solvers." *ACL 2020* (2020).
6. Koncel-Kedziorski, Rik, et al. "MAWPS: A math word problem repository." *NAACL 2016*.
7. Ling, Wang, et al. "Program induction by rationale generation: Learning to solve and explain algebraic word problems." *arXiv preprint arXiv:1705.04146* (2017).
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11. Wang, Xuezhi, et al. "Rationale-Augmented Ensembles in Language Models." *arXiv preprint arXiv:2207.00747* (2022).

Discussion

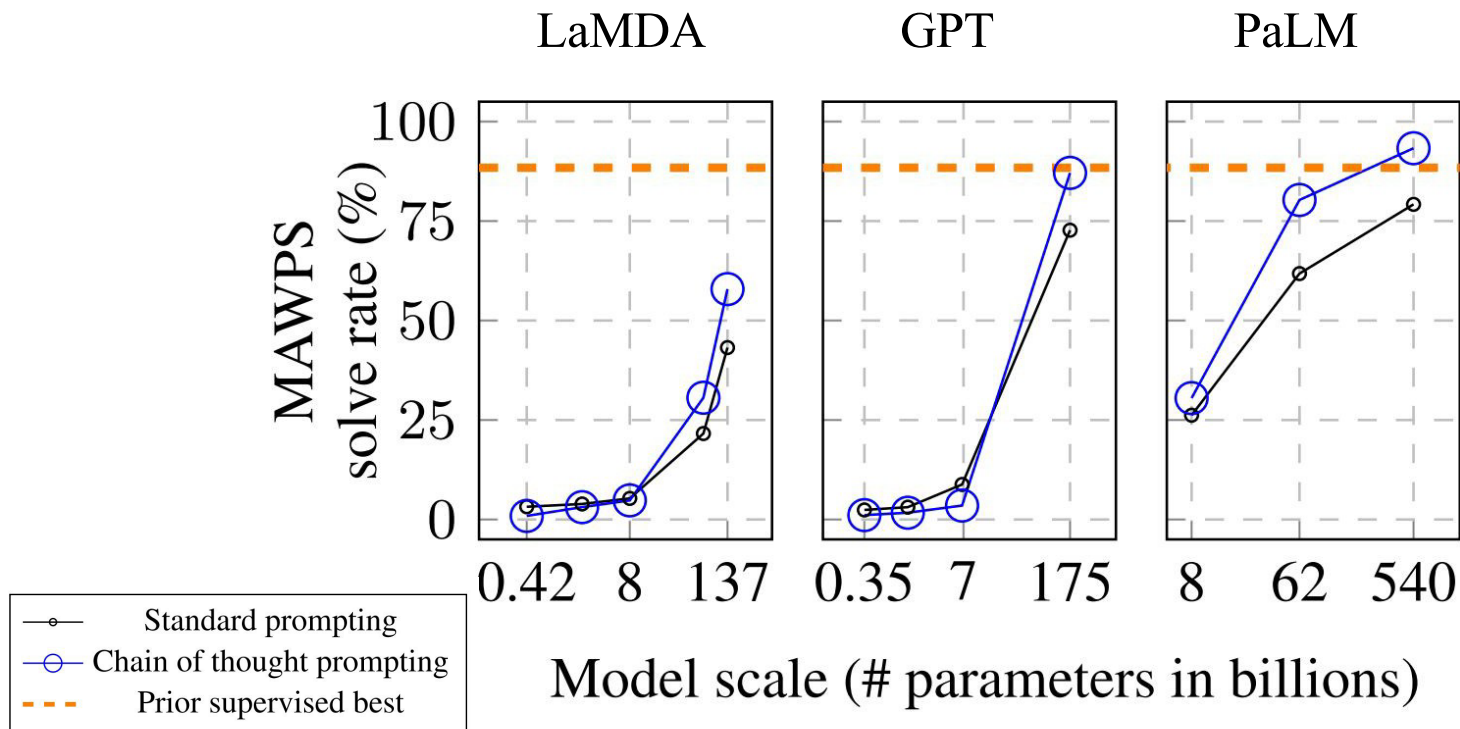
(This part will not be in presentation)

Summary of Arithmetic Reasoning Benchmark

Dataset	N	Example problem
GSM8K	1,319	Josh decides to try flipping a house. He buys a house for \$80,000 and then puts in \$50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?
SVAMP	1,000	Each pack of dvds costs 76 dollars. If there is a discount of 25 dollars on each pack. How much do you have to pay to buy each pack?
ASDiv	2,096	Ellen has six more balls than Marin. Marin has nine balls. How many balls does Ellen have?
AQuA	254	A car is being driven, in a straight line and at a uniform speed, towards the base of a vertical tower. The top of the tower is observed from the car and, in the process, it takes 10 minutes for the angle of elevation to change from 45° to 60° . After how much more time will this car reach the base of the tower? Answer Choices: (a) $5\sqrt{3} + 1$ (b) $6\sqrt{3} + \sqrt{2}$ (c) $7\sqrt{3} - 1$ (d) $8\sqrt{3} - 2$ (e) None of these
MAWPS: SingleOp	562	If there are 7 bottle caps in a box and Linda puts 7 more bottle caps inside, how many bottle caps are in the box?
MAWPS: SingleEq	508	Benny bought a soft drink for 2 dollars and 5 candy bars. He spent a total of 27 dollars. How much did each candy bar cost?
MAWPS: AddSub	395	There were 6 roses in the vase. Mary cut some roses from her flower garden. There are now 16 roses in the vase. How many roses did she cut?
MAWPS: MultiArith	600	The school cafeteria ordered 42 red apples and 7 green apples for students lunches. But, if only 9 students wanted fruit, how many extra did the cafeteria end up with?

Summary of math arithmetic reasoning benchmarks. N : number of evaluation examples (Wei, *et al.* 2022)

Arithmetic Reasoning - Few-Shot CoT MAWPS Results

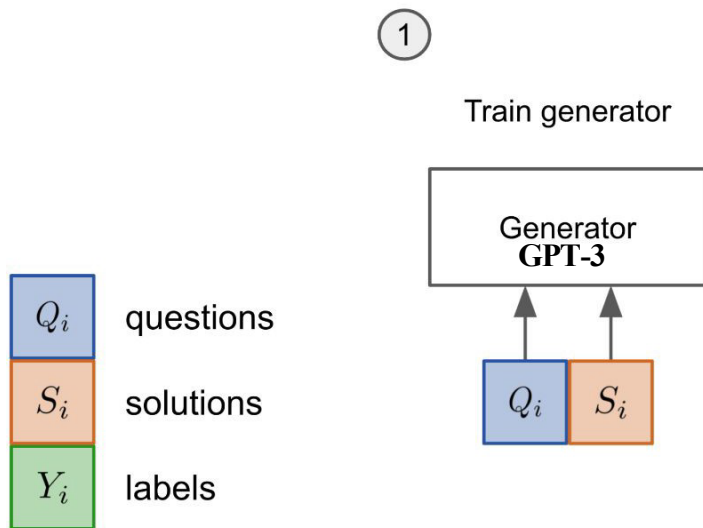


Arithmetic Reasoning - Zero-Shot CoT Additional Results

Table 3: Model scale study. Evaluation metric is accuracy on MultiArith dataset. 175B-1 : text-davinci-001, 175B-2 : text-davinci-002. It is verified that CoT is effective when the model is larger, such as Instruct GPT-3 (175B parameters; text-davinci-001 and text-davinci-002) and Original GPT-3 (175B parameters; davinci). In this experiment, the order of performance (ascending) is Zero-shot, Few-shot (8samples), Zero-shot-CoT, and Few-shot-CoT (8samples) for davinci and text-davinci-002.

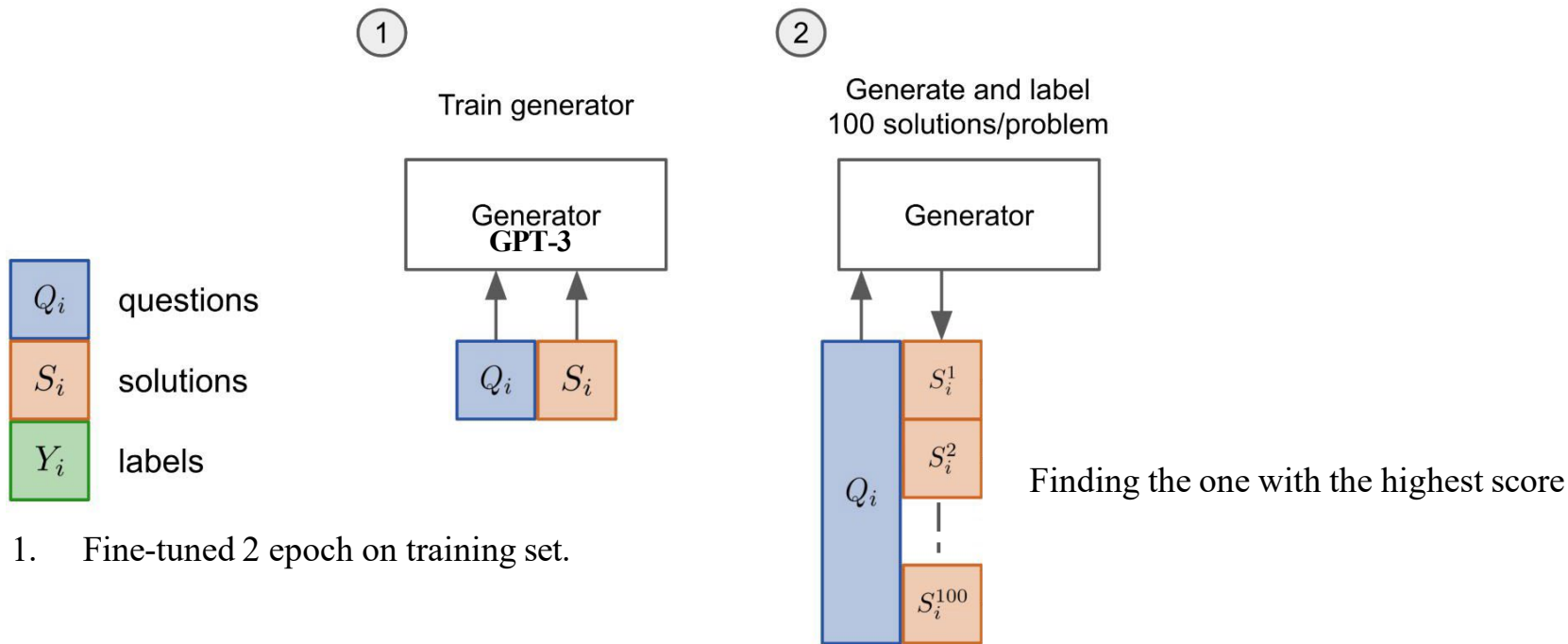
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Prior Best – Fine-tuning + Verification

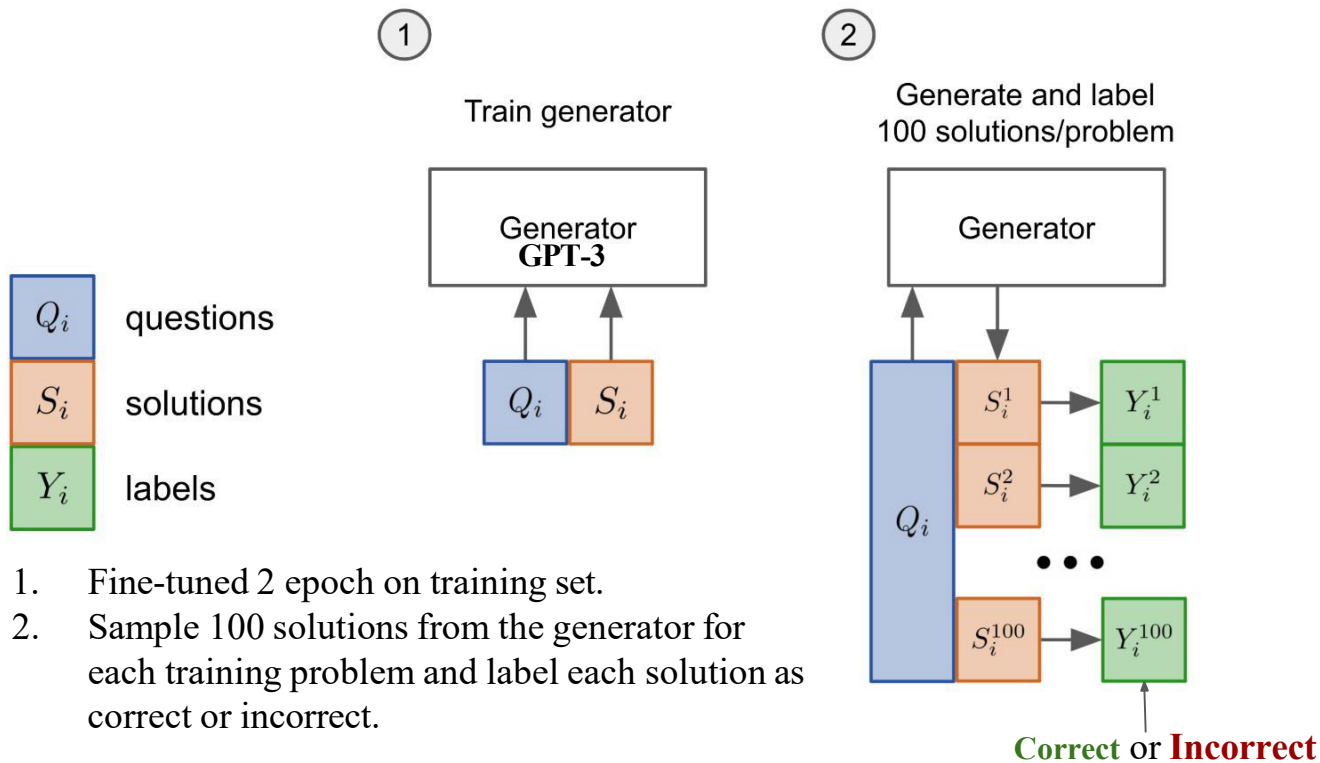


1. Fine-tuned 2 epoch on training set.

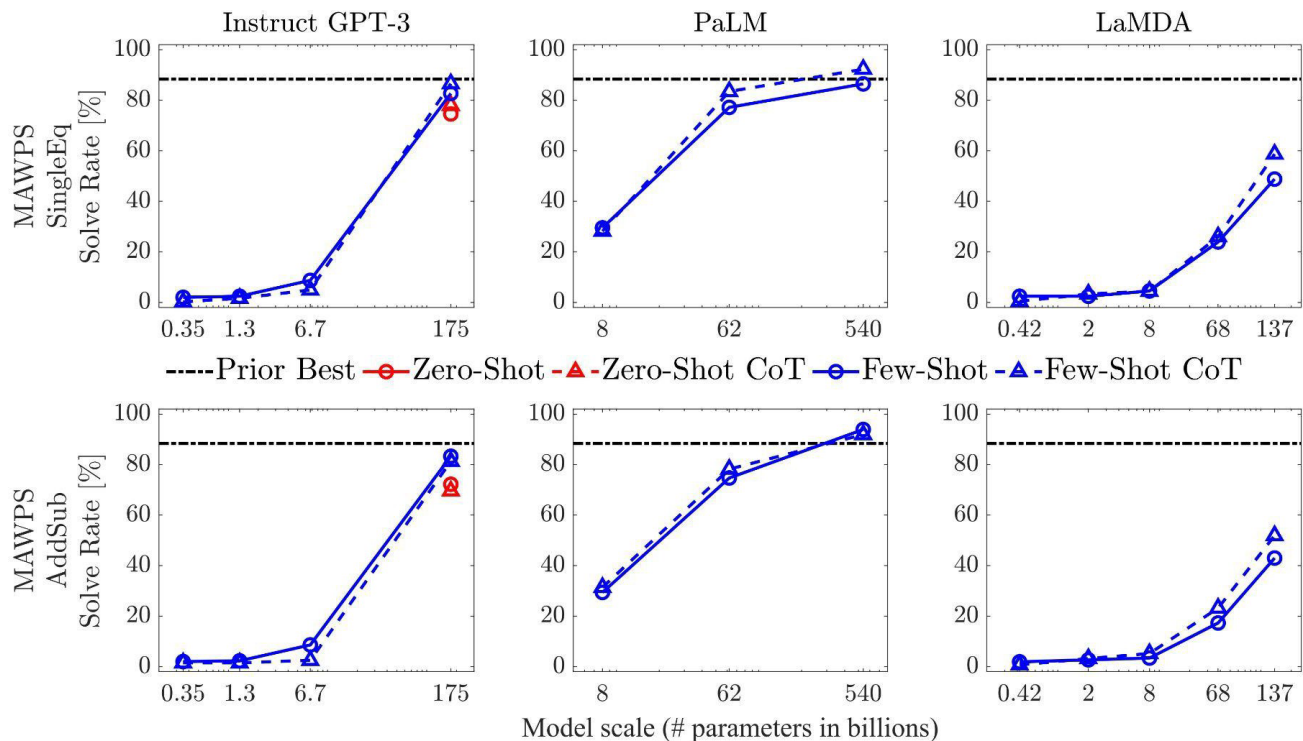
Prior Best – Fine-tuning + Verification



Prior Best – Fine-tuning + Verification



Arithmetic Reasoning - Results



MAWPS - SingleEq

If there are 7 bottle caps in a box and Linda puts 7 more bottle caps inside, how many bottle caps are in the box?

MAWPS - AddSub

There were 6 roses in the vase. Mary cut some roses from her flower garden. There are now 16 roses in the vase. How many roses did she cut?

Commonsense Reasoning - CSQA CoT Prompt

CSQA (Talmor et al., 2019)

Q: What home entertainment equipment requires cable?

Answer Choices: (a) radio shack (b) substation (c) television (d) cabinet

A: **The answer must require cable. Of the above choices, only television requires cable.** The answer is **(c)**.

- 7 exemplars from training dataset

Commonsense Reasoning - CSQA CoT Prompt

CSQA (Talmor et al., 2019)

Q: What home entertainment equipment requires cable?

Answer Choices: (a) radio shack (b) substation (c) television (d) cabinet

A: **The answer must require cable. Of the above choices, only television requires cable.** The answer is **(c)**.

- 7 exemplars from training dataset
- Manually composed intermediate reasoning with strict format:

The answer must _____. Of the above choices, only _____.

Commonsense Reasoning - Strategy QA CoT Prompt

StrategyQA (Geva et al., 2021)

Q: Could Brooke Shields succeed at University of Pennsylvania?

A: **Brooke Shields went to Princeton University. Princeton University is about as academically rigorous as the University of Pennsylvania. Thus, Brooke Shields could also succeed at the University of Pennsylvania.** The answer is **yes**.

- 6 exemplars from training dataset

Commonsense Reasoning - Strategy QA CoT Prompt

StrategyQA (Geva et al., 2021)

Q: Could Brooke Shields succeed at University of Pennsylvania?

A: **Brooke Shields went to Princeton University. Princeton University is about as academically rigorous as the University of Pennsylvania. Thus, Brooke Shields could also succeed at the University of Pennsylvania.** The answer is **yes**.

- 6 exemplars from training dataset
- Manually composed intermediate reasoning with flexible format

Commonsense Reasoning - Strategy QA CoT Prompt

StrategyQA (Geva et al., 2021)

Q: Could Brooke Shields succeed at University of Pennsylvania?

A: **Brooke Shields went to Princeton University. Princeton University is about as academically rigorous as the University of Pennsylvania. Thus, Brooke Shields could also succeed at the University of Pennsylvania.** The answer is **yes**.

StrategyQA (Geva et al., 2021)

Q: Yes or no: Is it common to see frost during some college commencements?

A: **College commencement ceremonies can happen in December, May, and June. December is in the winter, so there can be frost. Thus, there could be frost at some commencements.** The answer is **yes**.

Commonsense Reasoning - CoT Prompt

Date Understanding

Q: 2015 is coming in 36 hours.
What is the date one week from
today in MM/DD/YYYY

A: If 2015 is coming in 36 hours,
then it is coming in 2 days. 2
days before 01/01/2015 is
12/30/2014, so today is
12/30/2014. So one week from
today will be 01/05/2015. So the
answer is **01/05/2015**.

6 exemplars for Date Understanding

Sport Understanding

Q: Is the following sentence
plausible? "Jamel Murray was
perfect from the line."

A: **Jamal Murray is a basketball
player. Being perfect from the line
is part of basketball.** The answer
is **yes**.

8 exemplars for Sport Understanding

Commonsense Reasoning - CoT Prompt

SayCan Robot Planning (Ahn et al., 2022)

Locations = [counter, table, user, trash, bowl].

Objects = [7up, apple, kettle chips, tea, multigrain chips, coke, lime soda, jalapeno chips, rice chips, orange, grapefruit soda, pepsi, redbull, energy bar, sponge, water].

The robot can pick up items with **pick(object)** and put down items with **put(object)** as well as find objects or locations with **find()**. The robot can only understand the explicit locations and objects listed.

Human: How would you throw away a redbull?

Explanation: The user has asked me to throw away the redbull, I will move it to the trash.

Plan: 1. find(redbull), 2. pick(redbull), 3. find(trash), 4. put(redbull), 5. done().

Ablation Study - Robustness (Example Distribution)

Change the examples in **few-shot CoT**:

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are $16 / 2 = 8$ golf balls. Half of the golf balls are blue. So there are $8 / 2 = 4$ blue golf balls. The answer is 4. ✓

Change examples from in-domain to out-of-domain!

	Zero-shot	Few-shot-CoT [†]	Zero-shot-CoT	Few-shot-CoT
AQUA-RAT	22.4	<u>31.9</u>	33.5	39.0
MultiArith	17.7	<u>27.0</u>	78.7	88.2

Results for **few-shot** on two **AR** tasks with examples[†] from a CR task (CommonsenseQA): across-domain examples with the same format have minor performance degradation.